

Advanced Algebra: Concept Checklist

Active-Recall & Self-Testing Revision Guide

1 Number Systems, Decimal Expansions, and Algebraic Numbers

▮ Definitions & Concepts to State 1.1

- **Natural Numbers & Base Conversion:** Define natural numbers $\mathbb{N}$. Outline the algorithm to convert an integer between base 10 and an arbitrary base n .
- **Decimal Representations:** Distinguish between finite decimal expansions, pure repeating decimals, and eventually repeating decimals.
- **Algebraic vs. Transcendental Numbers:** Define what it means for a number $\alpha \in \mathbb{C}$ to be an *algebraic number* (root of a non-zero polynomial with rational coefficients) versus a *transcendental number*.
- **Quaternion Algebra ($\mathbb{H}$):** Define the skew field of quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ and state the fundamental multiplication rules:

$$i^2 = j^2 = k^2 = ijk = -1, \quad ij = k = -ji, \quad jk = i = -kj, \quad ki = j = -ik.$$

▮ Theorems & Identities to State 1.2

- **Finite Decimal Expansion Characterization:** A rational number in lowest terms $\frac{p}{q}$ ($\gcd(p, q) = 1$) has a finite decimal expansion $\Leftrightarrow$ the denominator q has no prime factors other than 2 or 5 (i.e., $q = 2^a 5^b$ for non-negative integers a, b).
- **Rational Decimal Representation Theorem:** A real number x is rational ($\mathbb{Q}$) $\Leftrightarrow$ its decimal expansion is finite or eventually periodic (repeating).
- **Algebraic Number Field Properties:** If α and β are algebraic numbers, then $\alpha + \beta$, $\alpha - \beta$, $\alpha\beta$, and α/β ($\beta \neq 0$) are all algebraic numbers.
- **Fundamental Theorem of Algebra:** Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_1 z + a_0$ with complex coefficients ($a_n \neq 0$) has at least one complex root, and exactly n roots in $\mathbb{C}$ when counted with multiplicity.

▮▮Proof Challenges & Calculations 1.3

- **Finite Decimal Proof:** Prove that $\frac{p}{q}$ (with $\gcd(p, q) = 1$) has a finite decimal expansion if and only if $q = 2^a 5^b$.
- **Repeating Decimal Characterization Proof:** Prove that every rational number $\frac{p}{q}$ can be written as a finite or repeating decimal using the Pigeonhole Principle on remainders during long division.

▮▮Textbook Exercise Checklist.

Section	Textbook Problems	Status
Chapter 1	Problem 1 (Section 1.1), Problems 4, 5, 6 (Section 1.3)	▫ Done

2 Polynomial Algebra, Division Algorithm, and Interpolation

▮ Definitions & Concepts to State 2.1

- **Polynomial Degree Properties:** Let $f(x), g(x) \neq 0$ with $\deg(f) = m$ and $\deg(g) = n$.
 - $\deg(f + g) \leq \max\{m, n\}$ (with strict equality if $m \neq n$).
 - $\deg(fg) = m + n$.
- **Polynomial Division Algorithm:** State the formal existence and uniqueness theorem: for polynomials $f(x)$ and $g(x) \neq 0$, there exist unique polynomials $q(x)$ (quotient) and $r(x)$ (remainder) such that:

$$f(x) = q(x)g(x) + r(x), \quad \text{where } \deg(r) < \deg(g) \text{ or } r(x) = 0.$$

- **Polynomial Greatest Common Divisor (gcd):** Define $\gcd(f(x), g(x))$ as the monic polynomial of maximum degree dividing both $f(x)$ and $g(x)$.
- **Lagrange Interpolation Formula:** State the explicit formula for the unique polynomial $f(x)$ of degree at most n passing through $n + 1$ distinct points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$:

$$f(x) = \sum_{i=0}^n y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

□ Theorems & Identities to State 2.2

- **Rational Root Theorem (Rational Factor Theorem):** Let $p(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients. If $ax - b$ is a linear factor of $p(x)$ with $\gcd(a, b) = 1$, then $a \mid a_n$ and $b \mid a_0$.
- **Euclidean Reduction for Polynomials:** If $f(x) = q(x)g(x) + r(x)$, then $\gcd(f(x), g(x)) = \gcd(g(x), r(x))$.
- **Bézout's Identity for Polynomials:** For any non-zero polynomials $f(x), g(x)$, there exist polynomials $a(x), b(x)$ such that:

$$a(x)f(x) + b(x)g(x) = \gcd(f(x), g(x)).$$

- **Euclid's Lemma for Polynomials:** If $f(x) \mid g(x)h(x)$ and $\gcd(f(x), g(x)) = 1$, then $f(x) \mid h(x)$.
- **Bounded-Degree Decomposition Theorem:** If $\gcd(f(x), g(x)) = 1$ and $h(x)$ is any polynomial with $\deg(h) < \deg(f) + \deg(g)$, there exist **unique** polynomials $a(x)$ and $b(x)$ with $\deg(a) < \deg(g)$ and $\deg(b) < \deg(f)$ such that:

$$h(x) = a(x)f(x) + b(x)g(x).$$

□ Proof Challenges & Calculations 2.3

- **Polynomial Division Uniqueness Proof:** Prove the uniqueness of the quotient $q(x)$ and remainder $r(x)$ in the polynomial division algorithm.
- **Rational Root Theorem Proof:** Prove the Rational Root Theorem for integer polynomials.
- **High-Degree Remainder Calculation:** Find the exact remainder when $x^{99} + x^2 - 1$ is divided by $x^2 - x + 1$.
- **Lagrange Interpolation Construction:** Construct a polynomial $f(x)$ of degree at most 5 satisfying:

$$f(-2) = 69, \quad f(-1) = 7, \quad f(0) = 1, \quad f(1) = 3, \quad f(2) = 11, \quad f(3) = 101.$$

▮ **Textbook Exercise Checklist.**

Section	Textbook Problems	Status
Section 2.2	Problems 2, 4, 9, 10, 11	□ Done
Section 2.3	Problems 3, 4, 6, 7, 8	□ Done
Section 2.4	Problems 6, 7	□ Done
Section 2.5	Problem 4	□ Done

3 Limits, Generalized Binomial Series, and Partial Fractions

▮ **Definitions & Concepts to State 3.1**

- **Generalized Binomial Theorem:** State the binomial series expansion for $(1+x)^\alpha$ where $\alpha \in \mathbb{R}$:

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k \quad \text{for } |x| < 1, \quad \text{where } \binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}.$$

- **Partial Fraction Decomposition Templates:** Outline the partial fraction decomposition setup for rational functions $P(x)/Q(x)$ ($\deg P < \deg Q$):
 - Distinct linear factors $(x-a)$: $\frac{A}{x-a}$.
 - Repeated linear factors $(x-a)^m$: $\sum_{j=1}^m \frac{A_j}{(x-a)^j}$.
 - Irreducible quadratic factors (x^2+px+q) : $\frac{Bx+C}{x^2+px+q}$.

▮ **Theorems & Identities to State 3.2**

- **Polynomial vs. Exponential Dominance Limit:** For any real exponent $k \in \mathbb{R}$ and any base $|x| < 1$:

$$\lim_{n \rightarrow \infty} n^k x^n = 0.$$

- **Geometric Series Identity:** For $|x| < 1$, $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$.

▮ Proof Challenges & Calculations 3.3

- **Partial Fraction Decompositions:** State the algebraic procedures to solve for unknown coefficients (Heaviside cover-up method and system of linear equations).
- **Binomial Series Expansion:** Expand $(1 - 2x)^{-1/2}$ up to the x^4 term using the generalized binomial coefficient formula.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 3.1	Problems 2, 7c, 11, 12	☐ Done
Section 3.2	Problems 4, 5	☐ Done
Section 3.3	Problems 6, 8, 9	☐ Done
Section 4.1	Problems 1, 3	☐ Done
Section 4.2	Problem 15	☐ Done

4 Complex Numbers, De Moivre's Theorem, and Polynomial Roots

Definitions & Concepts to State 4.1

- **Complex Number Fundamentals:** For $z = a + bi \in \mathbb{C}$ ($a, b \in \mathbb{R}$):
 - Define complex conjugate $\bar{z} = a - bi$.
 - Define modulus $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$.
 - Define argument $\arg(z)$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$.
 - Define trigonometric (polar) form $z = r(\cos \theta + i \sin \theta)$ and exponential form $z = re^{i\theta}$.

- **Euler's Formula & De Moivre's Theorem:** State Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ and De Moivre's identity:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) \quad (\forall n \in \mathbb{Z}).$$

- **Planar Rotation via Complex Numbers:** State the complex coordinate formula for rotating point $P(z_P)$ counter-clockwise by angle θ about center $A(z_A)$:

$$z' - z_A = e^{i\theta}(z_P - z_A).$$

- **n -th Roots of a Complex Number:** State the formula for all n distinct n -th roots of $z = re^{i\theta}$:

$$w_k = r^{1/n} \exp\left(i \frac{\theta + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1.$$

- **Vieta's Formulas:** For polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0$ with roots $z_1, \dots, z_n$, express the elementary symmetric sums $\sum z_i, \sum z_i z_j, \dots, \prod z_i$ in terms of coefficients a_k .

▮ Theorems & Identities to State 4.2

- **Modulus-Conjugate Identity:** $|z|^2 = z\bar{z}$, $|\bar{z}| = |z|$, and $|z_1 z_2| = |z_1||z_2|$.
- **Complex Conjugate Root Theorem:** Let $f(x) \in \mathbb{R}[x]$ be a polynomial with **real** coefficients. If $z \in \mathbb{C}$ is a root ($f(z) = 0$), then its complex conjugate $\bar{z}$ is also a root ($f(\bar{z}) = 0$).
- **Fundamental Theorem of Algebra:** Every degree- n complex polynomial factors completely into n linear factors over $\mathbb{C}$: $P(z) = a_n \prod_{k=1}^n (z - z_k)$.

▮ Proof Challenges & Calculations 4.3

- **Multiple-Angle Trigonometric Derivations:** Derive closed-form formulas for $\cos(nx)$ and $\sin(nx)$ in terms of powers of $\cos x$ and $\sin x$ by expanding $(\cos x + i \sin x)^n$ via Binomial Theorem.
- **Complex Conjugate Root Proof:** Prove that if $f(x) = \sum_{k=0}^n a_k x^k$ with $a_k \in \mathbb{R}$ and $f(z) = 0$, then $f(\bar{z}) = 0$.
- **Roots of Complex Numbers Calculation:**
 1. Find all 5 distinct roots of $w^5 = -32i$.
 2. Represent all 5 roots geometrically on the Argand diagram (complex plane).

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 5.3	Problems 6, 7, 9, 13, 14, 15, 16	☐ Done
Section 5.4	Problems 3, 5, 9, 12, 14, 15	☐ Done

5 Analytic Geometry, Axis Transformations, and Conic Sections

▮ Conventions & Exam Pitfalls.

- **Conic Discriminant Classification:** For the general quadratic curve $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:
 - $B^2 - 4AC < 0 \implies$ Ellipse (or circle / degenerate point / empty set).
 - $B^2 - 4AC = 0 \implies$ Parabola (or parallel lines).
 - $B^2 - 4AC > 0 \implies$ Hyperbola (or intersecting lines).
- **Rotation Angle Formula:** To eliminate the cross-product term Bxy , rotate through an angle θ satisfying $\cot(2\theta) = \frac{A-C}{B}$.

Definitions & Concepts to State 5.1

□ **Coordinate Transformations:**

- Translation to new origin (h, k) : $x' = x - h, y' = y - k$.
- Counter-clockwise rotation by angle θ : $x = x' \cos \theta - y' \sin \theta, y = x' \sin \theta + y' \cos \theta$.

□ **Parametric Plane Curves:** Define a plane curve given by $x = f(t), y = g(t)$ for $t \in [a, b]$. Define initial point, terminal point, closed curve, and simple curve.

□ **Weierstrass Universal Substitution:** State the half-angle parameterization formulas for $t = \tan(\theta/2)$:

$$\sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}, \quad d\theta = \frac{2 dt}{1+t^2}.$$

□ **Conic Eccentricity Definition:** Define a conic section by the ratio $e = \frac{|PF|}{|PL|}$ (distance to focus F over distance to directrix line L). Classify by eccentricity value ($e = 0, 0 < e < 1, e = 1, e > 1$).

□ **Standard Equations of Conics:** Write standard equations, foci, directrix, vertices, center, and asymptotes for:

- Parabola ($y^2 = 4px$ or $x^2 = 4py$)
- Ellipse ($\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, c^2 = a^2 - b^2, e = c/a$)
- Hyperbola ($\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, c^2 = a^2 + b^2, e = c/a$, asymptotes $y = \pm \frac{b}{a}x$)

□ **Tangent Line Formulas:** State the tangent line equation to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at point (x_0, y_0) : $\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$.

Theorems & Identities to State 5.2

- **Focal Distance Sum for Ellipse:** For any point P on an ellipse with foci F_1, F_2 , $|PF_1| + |PF_2| = 2a$.
- **Focal Distance Difference for Hyperbola:** For any point P on a hyperbola with foci F_1, F_2 , $||PF_1| - |PF_2|| = 2a$.
- **Conic Invariant Theorem:** The quantity $B^2 - 4AC$ is invariant under rigid rotations and translations of the coordinate axes.

▮ Proof Challenges & Calculations 5.3

- **Axis Translation Calculation:** Find the new coordinates of $P(6, 5)$ after a translation of axes to a new origin at $(2, 4)$.
- **Weierstrass Half-Angle Derivation:** Derive the formulas $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$ where $t = \tan(\theta/2)$ using double-angle trigonometric identities.
- **Tangent Line Derivation:** Derive the tangent line equation $\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1$ for an ellipse at (x_0, y_0) using implicit differentiation.

▮ Textbook Exercise Checklist.

Source & Section	Textbook Problems	Status
Varberg Sec 10.1	Problems 15, 30, 32, 37, 38, 39	▫ Done
Varberg Sec 10.2	Problems 43, 51	▫ Done
Varberg Sec 10.3	Problems 29, 35, 39, 41, 42, 46, 67	▫ Done
Varberg Sec 10.4	Problem 58	▫ Done
Stewart Sec 10.5	Problems 33, 45, 48, 57	▫ Done

6 Polar Coordinates and Polar Curves

▮ Definitions & Concepts to State 6.1

- **Polar-Cartesian Conversion:** State the conversion equations between (r, θ) and (x, y) :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}.$$

- **Standard Polar Graphs:** Define and sketch the general forms of:
 - Limaçons: $r = a \pm b \cos \theta$ and $r = a \pm b \sin \theta$.
 - Cardioids: $r = a(1 \pm \cos \theta)$ and $r = a(1 \pm \sin \theta)$ ($a = b$).
 - Lemniscates: $r^2 = a^2 \cos(2\theta)$ and $r^2 = a^2 \sin(2\theta)$.
 - Rose curves: $r = a \cos(k\theta)$ and $r = a \sin(k\theta)$ (determine the number of petals for odd vs. even k).
 - Spiral of Archimedes: $r = a\theta$.
- **Polar Lines and Circles:** Write the polar equation of a line $r \cos(\theta - \alpha) = d$ and a circle passing through the pole $r = 2a \cos \theta$ or $r = 2a \sin \theta$.
- **Unified Polar Equation of Conics:** State the unified polar equation of a conic with a focus at the pole and directrix $x = \pm d$ or $y = \pm d$:

$$r = \frac{ed}{1 \pm e \cos \theta} \quad \text{or} \quad r = \frac{ed}{1 \pm e \sin \theta}.$$

▮ Theorems & Identities to State 6.2

- **Rose Curve Petal Count Theorem:**

$$\text{Number of petals of } r = a \cos(k\theta) = \begin{cases} k & \text{if } k \text{ is odd} \\ 2k & \text{if } k \text{ is even} \end{cases}$$

- **Polar Rotation Invariance:** Replacing θ by $\theta - \alpha$ rotates the entire polar graph counter-clockwise by an angle α . (Specifically, $\cos \theta \rightarrow \sin \theta$ represents a $+90^\circ$ rotation).

▮Proof Challenges & Calculations 6.3

- **Polar-Cartesian Transformations:** Convert $r = \frac{6}{2-3\cos\theta}$ to standard Cartesian form and identify the conic section.
- **Unified Conic Polar Derivation:** Derive $r = \frac{ed}{1+e\cos\theta}$ from the geometric eccentricity definition $|PF| = e|PL|$ with focus at the origin and directrix $x = d$.

▮Textbook Exercise Checklist.

Source & Section	Textbook Problems	Status
Stewart Sec 10.3	Problems 11, 18	☐ Done
Stewart Sec 10.6	Problems 1, 3, 5, 7, 13	☐ Done

7 Three-Dimensional Space, Vector Geometry, Lines, and Planes

▮ Definitions & Concepts to State 7.1

▮ 3D Metrics and Angles:

- Distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ and midpoint formula.
- Direction angles α, β, γ and direction cosines $(\cos \alpha, \cos \beta, \cos \gamma)$.
- Scalar projection $\text{comp}_b a = \frac{a \cdot b}{\|b\|}$ and vector projection $\text{proj}_b a = \frac{a \cdot b}{\|b\|^2} b$.

▮ Cross Product and Triple Products:

- Cross product determinant formula: $a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$.
- Scalar triple product: $a \cdot (b \times c) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.
- Vector triple product: $a \times (b \times c)$.

▮ Lines and Planes in 3D:

- Vector, parametric, and symmetric equations of a line through r_0 with direction vector $v = (a, b, c)$:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

- Point-normal equation of a plane through (x_0, y_0, z_0) with normal $n = (A, B, C)$:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

- ▮ **Skew Lines:** Define skew lines (non-parallel, non-intersecting lines in $\mathbb{R}^3$).

▮ Curvilinear Coordinates:

- Cylindrical coordinates (r, θ, z) : $x = r \cos \theta, y = r \sin \theta, z = z$.
- Spherical coordinates (ρ, θ, φ) : $x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$.

▮ Theorems & Identities to State 7.2

- **Direction Cosines Identity:** $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.

- **BAC-CAB Identity (Vector Triple Product):**

$$a \times (b \times c) = (a \cdot c)b - (a \cdot b)c.$$

- **Parallelepiped Volume Theorem:** The volume of the parallelepiped determined by vectors a, b, c is $V = |a \cdot (b \times c)|$.

- **Coplanarity Theorem:** Three vectors $a, b, c \in \mathbb{R}^3$ are coplanar $\Leftrightarrow a \cdot (b \times c) = 0 \Leftrightarrow \det([a \mid b \mid c]) = 0$.

- **Distance from Point to Plane:** The distance from $P_0(x_0, y_0, z_0)$ to the plane $Ax + By + Cz + D = 0$ is:

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}.$$

- **Distance Between Two Skew Lines:** The distance between lines $L_1(r_1 + tv_1)$ and $L_2(r_2 + sv_2)$ is:

$$d = \frac{|(r_2 - r_1) \cdot (v_1 \times v_2)|}{\|v_1 \times v_2\|}.$$

▮ Proof Challenges & Calculations 7.3

- **Coplanarity Characterizations:** Show that three vectors $a, b, c \in \mathbb{R}^3$ are coplanar using:

1. Vector geometry and scalar triple products ($a \cdot (b \times c) = 0$).
2. Linear algebra matrix rank and determinants ($\det([a \mid b \mid c]) = 0$).

- **Plane Intersection Symmetric Equations:** Find the symmetric equations of the line of intersection of the two planes $x + y - z = 2$ and $2x - y + 3z = 1$.

- **Angle Between Two Planes:** Calculate the acute angle between planes $x + 2y - 2z = 5$ and $3x - 4y + z = 2$ using their normal vectors.

- **Skew Lines Distance Calculation:** Find the shortest distance between skew lines $L_1 : \frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{1}$ and $L_2 : \frac{x+2}{1} = \frac{y-3}{-2} = \frac{z-1}{4}$.

▮ Textbook Exercise Checklist.

Source & Section	Textbook Problems	Status
Stewart Sec 12.1	Problem 15	☐ Done
Stewart Sec 12.5	Problems 31, 51	☐ Done